Republic of the Philippines
Sorsogon State University
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
Bulan Campus
Zone 8, Bulan, Sorsogon
Tel. No.; 056 311-0103; Email Address: cict@sorsu.edu.ph

BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Activity No. 1 | Week 2: Introduction to Computing (CO1) | Unit 2, Computing Career Pathways

Explore Career Pathways

Name: Ariana kate L. Balaoro **Section:** BSCS 1-1

Course/Program: BSCS 1-1 **Date:** 09/16/26

I. Objectives

At the end of this activity, you should be able to:

- Identify the major computing career pathways introduced in lecture and describe what each one involves.
- Choose one computing career pathway that best matches your interests and explain why.
- Map a realistic entry-to-expert roadmap for your chosen pathway, including representative job titles and key skills at each stage.
- Compare your current skills against your chosen pathway's requirements and identify a concrete plan to close the gap.

II. Materials Needed

- Your CC 101 Unit 2 lecture slides/notes on Computing Career Pathways
- This worksheet (printed copy, or the digital file to fill out and submit)
- Optional: internet access, for Part E (real job posting search)

III. Background

Computing is not just "programming." As introduced in the Unit 2 lecture, the field branches into several distinct career pathways, each with its own operational scope, required technical competencies, representative job titles, and professional standard or reference body: Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, Cybersecurity & Threat Defense, and Information Systems & Enterprise IT. Each pathway also has its own career roadmap — the entry-level, mid-level, senior-level, and expert/leadership roles a professional typically grows through. This activity asks you to review those five pathways, choose the one that interests you most, and map out what growing in that pathway could look like for you.

IV. Procedure

Part A – Recall the Five Pathways

1. Using your lecture notes or slides, complete Table 1 by writing the core operational scope and one to two representative job titles for each of the five computing career pathways.

Part B – Choose Your Pathway

2. From the five pathways, choose ONE that interests you the most.
3. In two to three sentences, explain why you chose it (e.g., skills you already enjoy, a subject you like, a job title that excites you).

Part C – Map Your Career Roadmap

4. If your chosen pathway is Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, or Cybersecurity & Threat Defense, use the career roadmap shown for it in the lecture. If

you chose Information Systems & Enterprise IT (no roadmap was shown in lecture), research typical entry-level, mid-level, senior-level, and expert/leadership roles for this pathway online.

5. Complete Table 2 with example job titles and key skills/responsibilities at each of the four career stages.

Part D – Skill Gap Self-Assessment

6. Looking at the essential technical competencies for your chosen pathway (Table 1 / lecture), list at least five in Table 3. For each, mark whether you already have some experience with it, and briefly note how you plan to learn it if you don't.

Part E – Real-World Job Search (Optional / Bonus)

7. Search a real job platform (e.g., LinkedIn, Indeed, JobStreet, Kalibrr) for one entry-level job posting that matches your chosen pathway. Record it in Table 4.

V. Data and Findings

Table 1. The Five Computing Career Pathways (Recap — Part A)

Pathway	Core Operational Scope	Representative Job Title(s)
Software Engineering	Designing, testing, and building applications and algorithm-driven software systems	Software Engineer, Backend Developer, Full-Stack Engineer
Systems & Cloud Infrastructure	Maintaining hardware provisioning, network virtualization, uptime, and deployment pipelines	Systems Administrator, DevOps Engineer, Site Reliability Engineer (SRE)
Data Science & AI Engineering	Extracting insights from structured/unstructured data and training predictive AI models	Data Scientist, Machine Learning Engineer, Data Engineer
Cybersecurity & Threat Defense	Safeguarding network perimeters, conducting penetration testing, and managing incident response	Information Security Analyst, Penetration Tester, SOC Specialist
Information Systems & Enterprise IT	Aligning business workflows with technology solutions and enterprise software architectures	Business Systems Analyst, IT Manager, Solution Architect

My Chosen Pathway (Part B)

I am most interested in:

Why I chose it:

I am most interested in software engineering. I choose it because, I want to learn how to design, build applications. Also, someday want to develop a useful software that can help in every day life. I want to develop a software that is essential to everyone.

Table 2. My Career Roadmap for My Chosen Pathway (Part C)

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Entry-Level	Coder	• Mastering Best Practices • Algorithm Basics • Code & Debug • Learning Frameworks
Mid-Level	Developer	• Implementing Complex Features • Algorithm Efficiency • Unit Testing • Code Reviews
Senior-Level	Software Engineer	• System Architecture Design • Scalability & Performance • Monitoring juniors • Full Stack Responsibility
Expert / Leadership	Software Architect	• Guiding Technical • Team Leadership • Architectural Overviews • Cross-functional Collaboration

Table 3. Skill Gap Self-Assessment (Part D)

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Programming	Yes	More practice and try some way of programming
Debugging	Yes	I needed more practice, I can learn it by checking and analyzing the error
Data structures	No	By practicing and understanding it
Git	No	Practice on how to use git
Software architecture	No	Learn, practice and understanding.
Software testing	No	Learn and practice testing

Table 4. Real Job Posting Found (Part E, Optional / Bonus)

Item	Details
Job Title	Software engineer
Company	Accenture in the Philippines
Platform / Source (e.g., LinkedIn, Indeed, JobStreet)	LinkedIn
Required Skills Listed (3–5)	• Programming • software development • problem-solving • teamwork

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Why did you choose this particular pathway over the other four?

My chosen pathway is software engineering. Because, I want to have a depth understanding in programming. I want to learn how to design, to solve problems and create a useful software that everyone can use and essential in any pace of industry.

2. Looking at your roadmap, which career stage excites you the most, and why?

The career stages that excites me the most is the senior level, because I can lead projects and create more advanced software. Also, for me it implies my effort and achievement through out the time of working and learning.

3. Which required skill or competency for this pathway do you already have, and which one will be hardest for you to develop?

I have a basic computer skills , Programming and debugging will be the hardest for me to develop.

4. How does mapping this roadmap change — or confirm — your plans for the rest of this course and your degree?

This roadmap motivates me to improve my programming skills and prepare me for my future plans.

VII. Scoring Guide

Criteria	Description	Points
Completeness	Tables 1–3 and the reflection are fully filled out (Table 4 is bonus).	25
Roadmap Accuracy & Depth	Roadmap and skills recorded match the lecture pathway (or credible outside research for IS & Enterprise IT).	25
Realization / Reflection	Answers are thoughtful, specific to the student's own choice, and well written.	30
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	20
Total		**100**

Checked by: **Score:** / 100

Instructor's Signature over Printed Name / Date